# Exercises learning task 18

2026 – 2027

**Exercise 9.96:** Consider the following frequency table of observations on the random variable $X$.

| Values | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Observed frequency | 24 | 30 | 31 | 11 | 4 |

(a) Based on these $100$ observations, is a Poisson distribution with a mean of $1.5$ an appropriate model? Perform a goodness-of-fit procedure with $\alpha = 0.05$.

(b) Calculate the $p$-value for this test.

**Exercise 9.97:** Let $X$ denote the number of flaws observed on a large coil of galvanized steel. Of $75$ coils inspected, the following data were observed for the values of $X$:

| Values | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Observed frequency | 1 | 11 | 8 | 13 | 11 | 12 | 10 | 9 |

(a) Does the assumption of the Poisson distribution seem appropriate as a probability model for these data? Use $\alpha = 0.01$.

(b) Calculate the $p$-value for this test.

**Exercise 9.99:** Consider the following frequency table of observations on the random variable $X$:

| Values | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Frequency | 4 | 14 | 10 | 20 | 2 |

(a) Based on these $50$ observations, is a binomial distribution with $n = 6$ and $p = 0.25$ an appropriate model? Perform a goodness-of-fit procedure with $\alpha = 0.05$.

**Exercise 9.144:** Consider the spot weld shear strength data in the following table:

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 5409 | 5431 | 5475 | 5442 | 5376 | 5388 | 5459 | 5422 | 5416 | 5435 |
| 5420 | 5430 | 5401 | 5446 | 5487 | 5416 | 5382 | 5357 | 5388 | 5457 |
| 5407 | 5469 | 5415 | 5377 | 5454 | 5375 | 5409 | 5459 | 5445 | 5429 |
| 5463 | 5408 | 5481 | 5452 | 5422 | 5354 | 5421 | 5406 | 5444 | 5466 |
| 5399 | 5391 | 5477 | 5447 | 5330 | 5473 | 5423 | 5441 | 5412 | 5384 |
| 5445 | 5436 | 5454 | 5453 | 5428 | 5419 | 5465 | 5427 | 5421 | 5396 |
| 5381 | 5425 | 5388 | 5388 | 5378 | 5481 | 5388 | 5440 | 5482 | 5406 |
| 5401 | 5411 | 5399 | 5431 | 5440 | 5413 | 5406 | 5341 | 5452 | 5420 |
| 5458 | 5485 | 5431 | 5416 | 5431 | 5390 | 5399 | 5435 | 5386 | 5462 |
| 5383 | 5401 | 5407 | 5385 | 5440 | 5422 | 5448 | 5366 | 5430 | 5417 |

Does the normal distribution seem to be a reasonable model for these data? Perform an appropriate goodness-of-fit test to answer this question.